

# Matija Stepanic

Senior Software Engineer · Full-Stack & AI-Native Development

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DOMOVINA.ai: <https://domovina.ai>

## SUMMARY

Software engineer with 20 years of professional experience (since 2006) and 200+ delivered projects. MSc in Computer Science (FER, University of Zagreb). Founder & CEO of ITalk Ltd., Toptal Verified Expert, certified FlutterFlow Expert since 2022 with 50+ FlutterFlow consulting projects. Currently building AI-native products end-to-end: the DOMOVINA media-AI platform (RAG in ClickHouse, a public MCP server, Flutter, self-hosted Supabase) and PINKA on-chain crowdfunding. 4,465 GitHub contributions in the last year across 110 public repositories, working open-source-first with agentic AI tooling — 25B+ tokens processed with Claude Code, fully documented in a sourced public usage record. GitHub's raw counter reads 37,098, but 32,633 of those are automated commits publishing an open transcript corpus, so the corrected figure is the one quoted here.

## EXPERIENCE

### Founder & Lead Engineer

DOMOVINA.tv / DOMOVINA.ai | 06/2025 – present | Zagreb, Croatia — remote

- Designed and built the full platform solo using agentic AI development: Flutter app (web, iOS, Android, TV), self-hosted Supabase backend behind Kong on Coolify, and a Cloudflare edge layer — 4,465 GitHub contributions in a year across the ecosystem's code, separate from the 32,633 automated commits that publish the open transcript corpus.
- Built a 12-step media-ingestion pipeline ([fetch.domovina.tv](https://fetch.domovina.tv)) that has processed 3,157 Croatian podcast episodes across 48 channels — ~2,990 hours of audio, 2,698 distinct speakers — into 144,294 semantic RAG chunks: Whisper.cpp on Metal and NVIDIA Canary on Colab GPUs for ASR, pyannote diarization, Gemini summarisation via multi-region Vertex AI rotation.
- Implemented RAG search over the archive — bge-m3 embeddings in ClickHouse (HNSW, ~10–50 ms) with PostgreSQL metadata — and a production MCP server ([mcp.domovina.ai](https://mcp.domovina.ai)) with OAuth, per-client rate limiting and an audit dashboard, usable from Claude, ChatGPT and Cursor.
- Made RevenueCat the single billing system of record for the app across iOS, macOS, Android, Android TV and the web (RevenueCat Web Billing): one entitlement, a webhook mirroring entitlement state into Supabase Postgres behind RLS as the cross-platform source of truth, and optimistic unlock on mobile.
- Built [pay.domovina.ai](https://pay.domovina.ai) — SEPA EPC QR, Croatian HUB3 PDF417 and EIP-681 barcodes from a single form, with Safe 2/3 multisig routing of Monerium EURE on Gnosis Chain — plus a self-hosted SMS gateway, a catalogue of 901 Croatian football clubs and an interactive map of Croatia.

### Co-Founder & Lead Engineer

PINKA Finance | 09/2025 – present | Remote

- Architected and shipped the bilingual (HR/EN) marketing site and campaign/checkout app: Next.js 14 static export served by Cloudflare Workers with D1, custom i18n engine with CLDR plural rules.
- Designed a Foundry smart-contract suite (factory + ERC20Permit token + crowdfund) on Gnosis Chain with four parallel investment paths — ERC677 transferAndCall, approve+invest, SEPA via Monerium webhook, Safe multisig — and shipped the app as a Safe App that also runs framed inside Safe{Wallet}.
- Independently verified a public on-chain fan-token lottery (modric-lottery-verify) with reproducible scripts over chain data.

### Founder, CEO & Senior Software Engineer

ITalk Ltd. | 01/2016 – present | Zagreb, Croatia

- Worked as a FlutterFlow expert/consultant on 50+ projects: custom Flutter features, Firebase Cloud Functions integrations, published reusable open-source FlutterFlow components.
- Built a FlutterFlow identity-document scanning widget on top of BlinkID native SDKs, adopted by 1,119 different FlutterFlow projects.
- Developed FirePixy — photo-booth & event photo crowdsourcing platform: Node.js serverless backend (Firebase), Angular PWA, WPF photo-booth app integrated with the Canon EOS SDK.
- Led a team of 9 building ScanShop, a self-checkout retail platform: architecture and backend with NestJS + Parse Server, Angular PWA, native iOS/Android apps.
- Consulted for Microblink: server-side OCR bridge with API-key management for their SaaS, and a JavaScript web component + desktop-to-mobile handoff flow for document scanning.
- Built the backend for Tempo, an EU-funds project progress-tracking application (NestJS, Parse Server, Firebase), and a static-site platform (Angular + Scully + Flamelink CMS) powering multiple production sites since 2020.

## Software Engineer

Freelance | 01/2006 – 01/2016 | Zagreb, Croatia

- Developed an internal CRM for membership and activities of a Croatian public organisation — in production since 2008 with ~1,500 active users.
- Built a custom CMS and website for a Croatian business news agency; co-founded and built the backend of Mobicon, a crowd-microphone application.
- Delivered 100+ commercial websites and internal tools (PHP, Java, Node.js, MySQL/PostgreSQL/MongoDB, AWS, Spring Boot, Yii).

## SELECTED PROJECTS

**DOMOVINA.tv / DOMOVINA.ai** (Flutter, Dart, Python, Node.js, TypeScript, Supabase, PostgreSQL, ClickHouse, Whisper, Gemini, Cloudflare Workers, D1, Coolify, RAG, MCP, RevenueCat, Safe)

AI-native Croatian media platform of 30+ public repositories, built end-to-end by one engineer. An automated pipeline (Whisper.cpp on Metal, NVIDIA Canary on Colab GPUs, pyannote diarization, Gemini via multi-region Vertex AI) has processed 3,157 podcast episodes across 48 channels into 144,294 semantic RAG chunks, searchable through a public MCP server and a Flutter app that ships to web (WASM), iOS, Android and Android TV.

**domovina-rag — RAG backend & public MCP server** (TypeScript, Python, ClickHouse, PostgreSQL, bge-m3, MCP, Cloudflare R2, Coolify, Docker)

Self-hosted semantic-search backend over a Croatian podcast corpus: 144,294 RAG chunks from 3,157 episodes across 48 channels (~2,990 hours of audio, 2,698 distinct speakers), embedded with bge-m3 and stored in ClickHouse (HNSW vector index, ~10–50 ms queries) alongside PostgreSQL as the transactional source of truth. Exposed worldwide through a production MCP server, so Claude Desktop, Claude.ai, ChatGPT and Cursor can query the corpus as a native tool.

**PINKA Finance** (Next.js 14, TypeScript, Solidity, Foundry, Safe, Gnosis Chain, Monerium EURE, Supabase, Cloudflare, Tailwind)

Group-fundraising platform bridging bank transfers and on-chain money. Campaign & checkout app (Next.js 14) where backers pay by scanning a SEPA EPC QR code or on-chain via Monerium EURE on Gnosis Chain; Foundry smart-contract suite (PinkaFactory, PinkaToken, PinkaCrowdfund) supports four parallel investment paths — ERC677 transferAndCall, approve+invest, SEPA bank transfer via Monerium webhook, and Safe multisig.

**flutter\_certilia** (Flutter, Dart, OAuth2, eID)

Open-source Flutter package for the Certilia Identity API — Croatian national eID (NIAS) authentication for Flutter apps, with documentation site. 110+ commits in the last year.

**FirePixy** (Node.js, Firebase, Angular, WPF, Canon EDSDK, Flutter)

Complete platform for photo booths and photo walls at weddings and community events: serverless Node.js backend on Firebase, Angular PWA for guests, WPF photo-booth application integrated with Canon EOS DSLR cameras and on-site printing, Flutter mobile apps.

## **ScanShop** (NestJS, Parse Server, Firebase, Angular, Swift, Kotlin)

Self-checkout native mobile application for retail stores. Defined the architecture and built the backend (NestJS + Parse Server + Firebase) while leading a team of nine across backend, Angular PWA and native iOS/Android apps. Contributed improvements back to the open-source Parse Dashboard project.

## **FlutterFlow BlinkID Widget** (FlutterFlow, Flutter, BlinkID, iOS, Android)

Custom FlutterFlow widget wrapping native iOS/Android identity-document scanning SDKs (BlinkID). Reads identity documents with the device camera and returns structured data into FlutterFlow action chains. Adopted by 1,119 different FlutterFlow projects.

## **SKILLS**

Languages: TypeScript / JavaScript, Dart, SQL, Bash, PHP, Python, Solidity, Java, Kotlin, Swift, HTML / CSS / SCSS

Frameworks & Runtimes: Node.js, Flutter, FlutterFlow, Next.js / React, NestJS, Angular, Parse Server

Cloud & Infrastructure: Cloudflare (Workers, D1, R2, Pages), Firebase / GCP, Supabase (self-hosted), Docker / Coolify, AWS, CI/CD (GitHub Actions), PostgreSQL / MySQL / MongoDB

AI Engineering: Agentic development (Claude Code), RAG pipelines, MCP servers, LLM APIs (Anthropic, OpenAI)

Web3 / Blockchain: EVM / Solidity, Safe smart accounts, On-chain verification

Other: Software architecture, Team leadership, OCR / identity documents (BlinkID), Payments (Stripe, stablecoins, SMS)

## **EDUCATION**

### **Master of Science in Computer Science**

University of Zagreb, Faculty of Electrical Engineering and Computing (FER), 2013 – 2015

### **Bachelor of Science in Computer Science**

University of Zagreb, Faculty of Electrical Engineering and Computing (FER), 2009 – 2013

## **CERTIFICATIONS**

- Toptal Verified Expert in Engineering, Toptal, 2022-02
- Certified FlutterFlow Expert, FlutterFlow, 2022-05

## **LANGUAGES**

Croatian: Native; English: Full professional proficiency